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Original Article

Preliminary study to identify CXCR4 inhibitors as potential therapeutic agents for Alzheimer's and Parkinson's diseases

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Abstract

Neurodegenerative disorders (NDDs) are known to exhibit genetic overlap and shared pathophysiology. This study aims to find the shared genetic architecture of Alzheimer's disease (AD) and Parkinson's disease (PD), two major age-related progressive neurodegenerative disorders. The gene expression profiles of GSE67333 (containing samples from AD patients) and GSE114517 (containing samples from PD patients) were retrieved from the Gene Expression Omnibus (GEO) functional genomics database managed by the National Center for Biotechnology Information. The web application GREIN (GEO RNA-seq Experiments Interactive Navigator) was used to identify differentially expressed genes (DEGs). A total of 617 DEGs (239 upregulated and 379 downregulated) were identified from the GSE67333 dataset. Likewise, 723 DEGs (378 upregulated and 344 downregulated) were identified from the GSE114517 dataset. The protein–protein interaction networks of the DEGs were constructed, and the top 50 hub genes were identified from the network of the respective dataset. Of the four common hub genes between two datasets, C-X-C chemokine receptor type 4 (CXCR4) was selected due to its gene expression signature profile and the same direction of differential expression between the two datasets. Mavorixafor was chosen as the reference drug due to its known inhibitory activity against CXCR4 and its ability to cross the blood–brain barrier. Molecular docking and molecular dynamics simulation of 51 molecules having structural similarity with Mavorixafor was performed to find two novel molecules, ZINC49067615 and ZINC103242147. This preliminary study might help predict molecular targets and diagnostic markers for treating Alzheimer's and Parkinson's diseases.

Insight Box
Our research substantiates the therapeutic relevance of CXCR4 inhibitors for the treatment of Alzheimer's and Parkinson's diseases. We would like to disclose the following insights about this study. We found common signatures between Alzheimer's and Parkinson's diseases at transcriptional levels by analyzing mRNA sequencing data. These signatures were used to identify putative therapeutic agents for these diseases through computational analysis. Thus, we proposed two novel compounds, ZINC49067615 and ZINC103242147, that were stable, showed a strong affinity with CXCR4, and exhibited good pharmacokinetic properties. The interaction of these compounds with major residues of CXCR4 has also been described.

Keywords: CXCR4; Alzheimer's disease; Parkinson's disease; neurodegenerative disorders; molecular docking; molecular dynamics simulation

INTRODUCTION

Neurodegenerative disorders are reported to share a common pathophysiology. Genome-wide association data have identified genetic overlap and revealed common biological pathways between neurodegenerative diseases, such as Alzheimer's disease (AD) and Parkinson's disease (PD) [1]. Discovering common genetic architecture among different neurodegenerative disorders may help determine underlying shared disease mechanisms and facilitate early diagnosis and treatment strategies. AD and PD are age-related progressive neurodegenerative disorders with an enormous emotional and economic impact on patients and caregivers. AD is characterized by intracellular neurofibrillary tangles (NFT) composed of aggregated hyperphosphorylated tau protein in the neurons and glial cells [2, 3] and extracellular amyloid plaques consisting of aggregated amyloid-β (Aβ) [4, 5]. Even though PD is conventionally characterized by the aggregation of α-synuclein (α-Syn), tau protein and NFT have also been reported to modify PD symptomatology and disease risk [6–8]. Besides their activity in the immune system, chemokines and their receptors are remarkably expressed in the central nervous system (CNS) and modulate cell migration and neurotransmission.

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The CXC motif chemokine receptor 4 (CXCR4) belongs to the G protein-coupled receptor (GPCR) protein superfamily. CXCR4 and its ligand CXCL12 are usually linked with hematopoiesis. However, CXCL12 has also been reported to be expressed in other tissues such as the brain, heart, kidney, lung, thymus, liver, and spleen. Likewise, CXCR4-mediated biological pathways are associated with crucial cellular processes like cell proliferation, transport, and stress resistance [9]. CXCR4 performs an array of regulatory functions in the immune system and neurodevelopment [10–12]. CXCR4 has been reported to modulate axon guidance and apoptosis via microglial activation and astroglial signaling [13]. CXCR4 has also been reported to be involved in cell cycle regulation via p53 and Rb proteins [14, 15]. CXCR4 and functionally related genes have been associated with increased risk for various age-associated neurodegenerative diseases, such as AD and PD. CXCR4 expression has been observed to be elevated in AD and PD patients as compared to controls [16, 17]. The multifaceted activity of CXCR4 is explained by the complex mechanisms regulating its biological functions, such as receptor crosstalk, isoforms of receptor and ligands, non-canonical ligands, and signaling break-off. Impairment of the CXCR4/CXCL12 signaling and consequent biological processes are associated with an array of disease pathologies, including neurodegenerative disorders, autoimmune diseases, immunodeficiency disorders, developmental abnormality, and malignancy [9].

Due to technological advancements, a large amount of genomic, transcriptomics, and proteomic information is now available for discovering new drug targets and screen new lead compounds [18]. Computer-aided drug discovery (CADD) is emerging as a powerful tool that employs this information to accelerate drug design through advanced computational methods and mathematical modeling to simulate receptor–drug interactions and predict binding affinities to screen potential compounds without prior purchase or synthesis, thereby saving significant amount of time and money [19]. CADD is usually classified into two categories: structure-based drug discovery (SBDD) and ligand-based drug discovery (LBDD). SBDD aims to design new lead compounds by analyzing 3D structure information of receptors (proteins) to determine binding sites and interactions vital for their biological activity. In comparison, LBDD utilizes physiochemical properties and biological activity of known ligands to optimize existing drugs or design new drugs with enhanced activity [20]. Some of the commonly utilized CADD methodologies are receptor (protein) identification, molecular docking and simulation studies, QSAR (quantitative structure activity relationship), absorption, distribution, metabolism, excretion and toxicity (ADMET) properties, molecule design, and lead optimization [21]. Molecular docking is the most frequently used SBDD methodology facilitated by exponential growth in the number of easily accessible 3D structures of receptors (proteins) and small molecule compounds (ligands), and availability of computation power to analyze them [22]. It is used to predict orientation and affinity of interaction between a ligand and a protein in its binding site, and provide a docking score that signifies the binding strength between ligand and target protein [23].

MATERIALS AND METHODS

Data sources

The datasets with accession numbers GSE67333 and GSE114517 (Fig. 1) were retrieved from the Gene Expression Omnibus (GEO) [24] database (https://www.ncbi.nlm.nih.gov/geo/). GSE67333 RNA-seq dataset contains four samples from late-onset Alzheimer's disease (LOAD) patients and four samples from age-matched healthy controls. The dataset is a comprehensive list of transcriptomics alterations and warrants a holistic approach, including both coding and non-coding RNAs in functional studies to understand the pathophysiology of LOAD. In addition, GSE114517 RNA-seq dataset contains 17 samples from Parkinson's disease patients and 13 samples from age-matched healthy controls. In GSE114517, for performing RNA-seq, the samples, acquired from the Netherland's Brain Bank, were taken post mortem from PD and non-PD control donors. For RNA extraction, tissue pieces were cut on dry ice and snap-frozen in liquid nitrogen to maintain RNA integrity. Lysis and homogenization (using a pellet pestle) were performed with 700 μl QIAzol Lysis Reagent, after which homogenates were snap-frozen again, before RNA extraction with the miRNeasy Kit.

Molecular docking analysis

The docking analyses were performed using AutoDock Vina [25]. The CXCR4 protein structure (PDB Code: 3ODU) was retrieved from Protein Data Bank (www.rcsb.org) [26]. The acquired CXCR protein has two chains with a sequence length of 502 and resolution of 2.50 Å. The structure reveals a consistent homodimer with an interface including helices V and VI. All undesired molecules (water molecules, ligands, and cofactors) were removed, and hydrogen atoms were added. The structure of the reference drug, Mavorixafor, was retrieved from PubChem (www.pubchem.ncbi.nlm.nih.gov/compound/amd-070). ZINC database [27] was used for structure similarity search, and SDF files of 51 molecules with more than 40% similarity were selected for further analysis (Fig. 1). Avogadro software [28] was used to convert the SDF format to the MOL2 format.

Prediction of pharmacokinetic/toxicity (ADMET) properties

Bioavailability radar

The bioavailability radar is a tool that considers six different physicochemical properties (size, solubility, lipophilicity, flexibility, polarity, and saturation) to predict a molecule's drug-likeness. Each axis of the radar represents one of these properties whose range is determined by different descriptors [29, 30] and depicted as a pink zone where the radar plot of the molecule must fall for it to be deemed as drug-like.

Physicochemical properties

The physicochemical properties consist of various physical and chemical descriptors like the number of particular atom types, the number of specific bond types, molecular weight (MW), fraction of sp3 hybridized carbons (Fraction Csp3), molecular refractivity (MR), and topological polar surface area (TPSA) and are utilized to predict absorption and blood–brain permeability. The PSA is calculated using the fragmental technique called TPSA, considering sulfur and phosphorus as polar atoms. This has proven to be a useful descriptor in many models and rules to quickly estimate some ADME properties, especially with regard to biological barrier crossing such as absorption and brain access [30, 31].

Lipophilicity

The lipophilicity of drugs is measured by evaluating the octanol/water partition coefficient (log $P_{o/w}$). Many computational methods for log $P_{o/w}$ estimation were developed with diverse performance on various chemical sets. Common practice is to use multiple predictors either to select the most accurate

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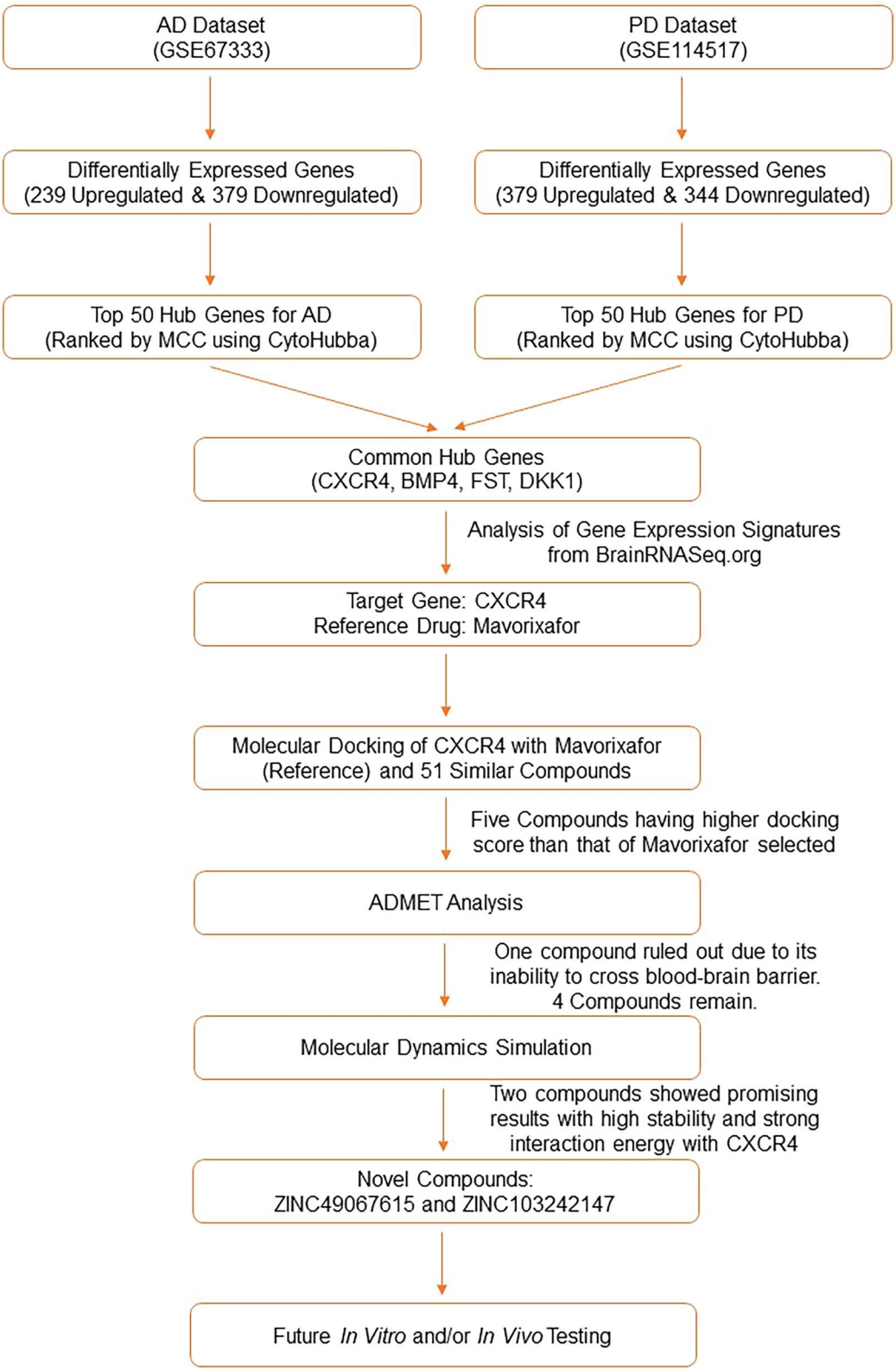


Figure 1. Methodology for virtual screening of CXCR4 inhibitors as potential therapeutic agents for Alzheimer's and Parkinson's diseases. The datasets with accession numbers GSE67333 and GSE114517 were retrieved from the Gene Expression Omnibus (GEO) database. To determine the differentially regulated gene (DEG) profiles for AD and PD patients, the datasets GSE67333 and GSE114517 were analyzed with the GEO RNA-seq Experiments Interactive Navigator (GREIN) web platform. STRING, a database of known and predicted interacting genes/proteins of a wide range of organisms, was searched for potential protein–protein interactions between the proteins encoded by the DEGs obtained from the datasets GSE67333 (AD) and GSE114517 (PD). Top 50 hub genes were screened separately from the datasets GSE67333 (AD) and GSE114517 (PD) using CytoHubba plugin for Cytoscape. Virtual screening was performed via molecular docking to screen novel potential therapeutic agents, by computing their binding affinity score and interacting residues of our target protein, CXCR4. 3D structure of selected protein CXCR4 (PDB ID: 3ODU) was retrieved from Protein Data Bank. Mavorixafor and all five test molecules were analyzed to predict their ADME properties. Five test molecule complexes with CXCR4 displaying the highest docking scores were chosen for molecular dynamics simulations to validate the molecular docking results.

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methods for a given chemical series or to generate consensus estimation. SwissADME calculates five predictors (iLOGP, MLOGP, WLOGP, XLOGP3, and SILICOS-IT) to compute the consensus log $P_{o/w}$ (average of all five predictors) [32, 33]. Here, XLOGP3 is an atomistic method including corrective factors and knowledge-based library29; WLOGP, our own implementation of a purely atomistic method based on the fragmental system of Wildman and Crippen; MLOGP, an archetype of topological method relying on a linear relationship with 13 descriptors; and SILICOS-IT, a hybrid method relying on 27 fragments and 7 descriptors.

Water solubility

The water solubility of molecules is calculated by SwissADME using three methods—ESOL (Estimated Solubility) [34], Ali [35], and SILICOS-IT—as log S values that are the decimal logarithm (i.e. logarithm with base 10) of their molar aqueous solubility and is calculated in moles per liter and milligrams per milliliter. Their qualitative solubility class is also provided.

Pharmacokinetics

SwissADME utilizes various specialized models to assess the ADME properties of molecules. The first model predicts the blood–brain barrier (BBB) permeability and passive gastrointestinal absorption [31]. The second model predicts whether a molecule is a substrate of permeability glycoprotein (Pgp). It assesses a molecule's ability to permeate through membranes such as the gastrointestinal wall and the brain [36]. The third model predicts the interaction of a molecule with major cytochrome P450 (CYP) isoforms (CYP1A2, CYP2C9, CYP2C19, CYP2D6, CYP3A4) that play a crucial role in drug elimination by utilizing biotransformation. Furthermore, their inhibition leads to drug interactions [37], resulting in toxicity or other undesired ramifications. The last model predicts the skin permeability coefficient (K_p) that shows a linear correlation between molecular mass and lipophilicity [38]. An increase in the negative value of log K_p (cm/s) corresponds to a decrease in skin permeability. SwissADME applied the support vector machine algorithm on meticulously cleansed large datasets of known substrates/non-substrates or inhibitors/non-inhibitors.

Drug-likeness

It qualitatively analyzes the potential of a molecule to act as a therapeutic agent regarding its oral bioavailability by comparing the physicochemical properties of a molecule with its biopharmaceutical nature inside the human body [39]. SwissADME utilizes multiple filters consisting of different ranges of descriptors that must be fulfilled for a molecule to be considered drug-like. There filters are Lipinski [40], Ghose [41], Veber [42], Egan [43], and Muegge [44]. In addition, a bioavailability score [45] is given to the molecules that describe them in four classes with oral bioavailability of 0.11, 0.17, 0.55, or 0.85. It seeks to predict the probability of a compound to have at least 10% oral bioavailability in rat or measurable Caco-2 permeability. Multiple estimations allow consensus views or selection of methods best fitting the end-user's specific needs in terms of chemical space or project-related demands. Any violation of any rule described here appears explicitly in the output panel.

Medicinal chemistry friendliness

SwissADME removes PAINS (pan assay interference compounds) using two pattern recognition filters. These compounds can potentially cause problems in assays irrespective of the target protein and can also lead to false-positive results [46]. The problematic functional groups or moieties, which can cause metabolic imbalance, toxicity, or inferior pharmacokinetics, are also displayed as 'Structural Alerts'. Lead-likeness can be similar to drug-likeness, but it focuses more on the physicochemical properties that must be fulfilled for a molecule to be regarded as a good lead [47]. The 'Synthetic Accessibility' score that lies between 1 (easy synthesis) and 10 (complex synthesis) provides an estimation of the ease of synthesis of drug-like molecules. It allows the selection of promising virtual screened molecules for synthesis and further biological testing [48].

Molecular dynamics simulations

The 50 ns molecular dynamics (MD) simulation was done for the docked reference and test molecule structures (Fig. 1). The best-docked structures were selected for 50 ns MD simulation using GROMACS package [49]. The protein and ligands were split into separate structure files, and their topology of protein was produced using CHARMM36 force field and default (TIP3P) water conditions. Ligand topology was created using the CGenFF web-server by feeding the Mol2 file for ligand after arranging the bond orders. The solvation was performed by defining the unit cell and filling it with a solvent like water. The system was neutralized by introducing the appropriate number of counter ions. The steepest descent method was utilized for energy minimization and subsequently positioned restraining the ligand. Later, the system was equilibrated in a constant volume (NVT) and constant pressure (NPT) environment for 100 ps each. V-Rescale temperature coupling was used to maintain 300 K temperature, while C-rescale pressure coupling was used to hold 1 bar pressure. Finally, the production MD was run after releasing the restraints for data collection. The produced trajectories were evaluated for root-mean-square deviations (RMSD), radius of gyration (Rg), root-mean-square fluctuation (RMSF) per residue, and interaction energies of Coulomb and Lennard–Jones.

RESULTS

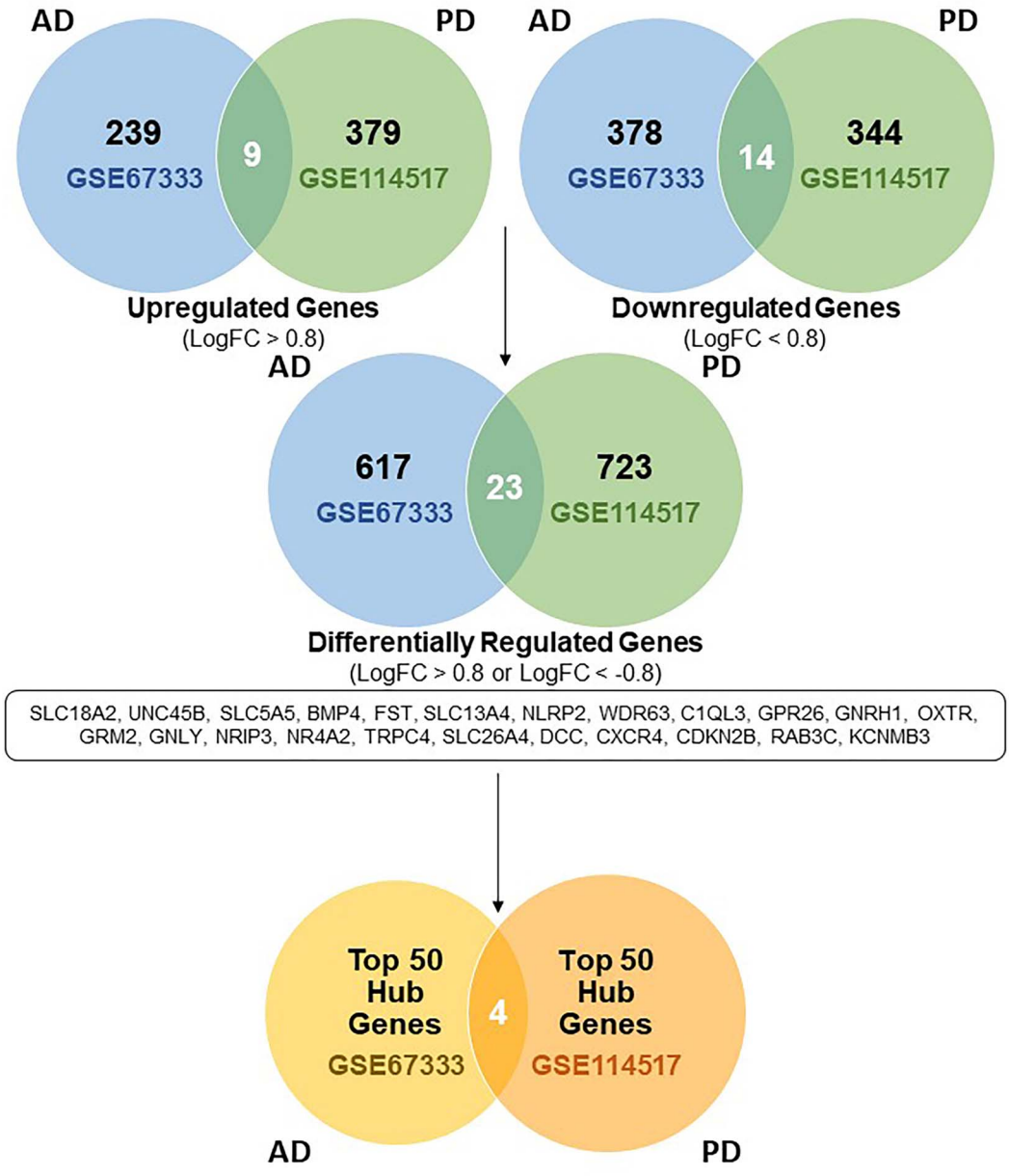
Differentially regulated genes concurrently expressed in AD and PD patients

The differential expression was measured by comparing expression level of genes in AD or PD patients with that of aged-matched healthy controls (in respective datasets). To acquire the list of significant DEGs from the individual datasets, $|\log_2 FC| > 0.8$ and $P < 0.05$ were taken as cutoff values, where DEGs were upregulated if $\log_2 FC > 0.8$ and downregulated if $\log_2 FC < -0.8$. From the GSE67333 dataset of AD patients and aged-matched healthy controls, 617 DEGs were identified, out of which 239 genes were upregulated ($P < 0.05$, $\log_2 FC > 0.8$) and 379 genes were downregulated ($P < 0.05$, $\log_2 FC < -0.8$) (Fig. 2). Likewise, from GSE114517 dataset of PD patients and aged-matched healthy controls, 723 DEGs were identified, out of which 378 genes were upregulated ($P < 0.05$, $\log_2 FC > 0.8$) and 344 genes were downregulated ($P < 0.05$, $\log_2 FC < -0.8$) (Fig. 2). Finally, to identify common DEGs between AD and PD, two datasets were run on Venn Diagram, and 23 common DEGs were confirmed, out of which 9 genes were upregulated ($P < 0.05$, $\log_2 FC > 0.8$) and 14 genes were downregulated ($P < 0.05$, $\log_2 FC < -0.8$) (Table 1 and Fig. 2).

Protein–protein interaction network and hub genes analysis

Protein–protein interactions play a crucial role in predicting target proteins' biological function in healthy and diseased individuals

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Gene Expression in Different Neuronal Cells (BrainRNASeq)

Genes	Astrocytes	Neurons	Oligodendrocytes	Microglia
CXCR4	**7.91**	**1.10**	**7.92**	**109.80**
BMP4	0.10	0.20	0.10	0.10
FST	0.10	0.10	0.10	0.10
DKK1	0.22	0.90	0.24	0.10

Figure 2. Analysis of differential gene expression. GEO RNA-seq Experiments Interactive Navigator (GREIN) web platform was used to find differentially regulated genes from datasets GSE67333 (AD) and GSE114517 (PD). To acquire the list of significant DEGs from the individual datasets, $|\log2FC| > 0.8$ and $P < 0.05$ were taken as cutoff values, where DEGs were upregulated if $\log2FC > 0.8$ and downregulated if $\log2FC < -0.8$. From GSE67333 Dataset, 617 DEGs were identified, 239 upregulated genes ($P < 0.05$, $\log2FC > 0.8$) and 379 downregulated genes ($P < 0.05$, $\log2FC < -0.8$). From GSE114517 Dataset, 723 DEGs were identified, 378 upregulated genes ($P < 0.05$, $\log2FC > 0.8$) and 344 downregulated genes ($P < 0.05$, $\log2FC < -0.8$). The top 50 Hub Genes were identified from both datasets, and common 4 hub genes were selected. To confirm expression of selected hub genes, BrainRNASeq was used, as it contains experimental RNA sequencing data of different brain cells, *viz*., astrocytes, neurons, oligodendrocytes, and microglia. A gene was considered to be significantly expressed if its FPKM value was more than 0.5. Only CXCR4 gene was significantly expressed (FPKM >0.5) in all cell types and had FPKM values of 7.91 (astrocytes), 1.10 (neurons), 7.92 (oligodendrocytes), and 109.80 (microglia). It was also upregulated in both AD and PD. Thus, CXCR4 was selected for the next part of the study.

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Table 1. Common differentially regulated genes (DEGs) between datasets GSE67333 and GSE114517.

Gene		GSE67333		GSE114517	
		LogFC	*P*-value	LogFC	*P*-value
Upregulated genes	Bone morphogenetic protein 4 (BMP4)	2.151	0.00023	1.206	0.00766
	Cyclin dependent kinase inhibitor 2B (CDKN2B)	0.882	0.01028	0.999	0.01369
	C-X-C motif chemokine receptor 4 (CXCR4)	0.958	0.03434	0.933	0.00854
	Gonadotropin-releasing hormone 1 (GNRH1)	0.986	0.01160	1.420	0.00231
	Glutamate metabotropic receptor 2 (GRM2)	1.222	0.00398	1.046	0.04115
	Potassium calcium-activated channel subfamily M regulatory beta subunit 3 (KCNMB3)	0.896	0.00805	0.903	0.04042
	Solute carrier family 13 member 4 (SLC13A4)	1.594	0.01058	1.304	0.00105
	Solute carrier family 26 member 4 (SLC26A4)	1.063	0.00123	0.928	0.04888
	Solute carrier family 5 member 5 (SLC5A5)	2.996	0.00010	1.050	0.01544
Downregulated genes	Complement C1q like 3 (C1QL3)	−0.970	0.01215	−1.626	0.02612
	DCC netrin 1 receptor (DCC)	−0.889	0.01781	−1.076	0.01732
	Follistatin (FST)	−1.311	0.00604	−1.939	0.03004
	Granulysin (GNLY)	−1.256	0.00473	−0.984	0.03550
	G protein-coupled receptor 26 (GPR26)	−1.087	0.00291	−1.348	0.02741
	NLR family pyrin domain containing 2 (NLRP2)	−0.933	0.00145	−2.197	0.00452
	Nuclear receptor subfamily 4 group A member (NR4A2)	−1.151	0.00624	−0.990	0.03180
	Nuclear receptor interacting protein 3 (NRIP3)	−1.208	0.00787	−0.990	0.01343
	Oxytocin receptor (OXTR)	−1.289	0.00787	−1.036	0.00387
	RAB3C, member RAS oncogene family (RAB3C)	−0.803	0.03211	−1.056	0.03173
	Solute carrier family 18 member A2 (SLC18A2)	−2.972	0.00002	−2.379	0.00806
	Transient receptor potential cation channel subfamily C member 4 (TRPC4)	−0.984	0.01605	−1.076	0.03033
	Unc-45 myosin chaperone B (UNC45B)	−2.166	0.00101	−1.463	0.00963
	WD repeat domain 63 (WDR63)	−1.698	0.00037	−1.099	0.01050

[50]. STRING (version 11.5) [51], a database of known and predicted interacting genes/proteins of a wide range of organisms, was searched for potential protein–protein interactions between the proteins encoded by the DEGs obtained from the datasets GSE67333 (AD) and GSE114517 (PD). These interactions can be either physical or functional in nature, and derived from co-expression analysis, text-mining of the scientific literature, or computational prediction. Significant interactions, with the minimum confidence score of 0.4, were selected to build PPI networks in order to identify the most important genes and proteins that may play a vital role in development of AD and PD, respectively. The resulting PPI networks were obtained from the STRING database and exported as short tabular text output (.tsv) files. Cytoscape (version 3.9.1) was used to analyze these PPI networks and identify the hub genes [52]. CytoHubba plugin for Cytoscape was used to determine top hub genes graded by Maximal Clique Centrality (MCC) score [53]. Given a node v, the MCC of v is defined as $MCC(v) = \sum_{C \in S(v)} (|C| - 1)!$, where $S(v)$ is the collection of maximal cliques that contain v, and $(|C| - 1)!$ is the product of all positive integers less than $|C|$. If there is no edge between the neighbors of the node v, then $MCC(v)$ is equal to its degree. Top 50 hub genes were screened separately from the datasets GSE67333 (AD) and GSE114517 (PD). Four common hub genes, CXCR4, DKK1, BMP4, and follistatin (FST), were found (Fig. 2) between the top 50 hub genes obtained from datasets GSE67333 (AD) and GSE114517 (PD). Studies have demonstrated that in AD and PD, CXCR4 modulates TLR4 signaling pathway that leads to MAPK activation and the production of inflammatory cytokines [54]. Furthermore, it also leads to inhibition of Wnt signaling, modulation of Tau phosphorylation, and induction of neuronal cell death by DKK1 [55]. Similarly, Dun *et al.* demonstrated that inhibition of canonical Wnt pathway by DKK contributes to the etiology of PD rat model [56]. In addition, studies have found the upregulation of TGFβ superfamily ligand, BMP4, in the central nervous system after KA-induced neurodegeneration [57]. FST is a secreted extracellular glycoprotein expressing widely in the nervous system. Downregulation of FST disrupts synaptic transmission in hippocampus and leads to cognitive impairments [58].

Analyzing gene expression signatures of screened genes in different cell types

To confirm expression of selected hub genes, BrainRNASeq was used, as it contains experimental RNA sequencing data of different brain cells, *viz.* astrocytes, neurons, oligodendrocytes, and microglia. In BrainRNASeq, single-cell RNA-Seq profiles for microglia and brain myeloid cells (1816 cells total passed quality control) from different developmental stages (E14.5, P7, and P60) were generated, which were grouped into 15 clusters using Seurat package. A gene was considered to be significantly expressed if its FPKM value was more than 0.5. BMP4 was not significantly expressed in any cell type and had FPKM values of 0.10 (astrocytes), 0.20 (neurons), 0.10 (oligodendrocytes), and 0.10 (microglia). FST was also not significantly expressed in any cell type and had FPKM values of 0.10 (astrocytes), 0.10 (neurons), 0.10 (oligodendrocytes), and 0.10 (microglia). DKK1 was not significantly expressed in any cell type besides neurons and had FPKM values of 0.22 (astrocytes), 0.90 (neurons), 0.24 (oligodendrocytes), and 0.10 (microglia). However, DKK1 was upregulated in AD while downregulated in PD and, therefore, was ruled out. The only remaining gene, CXCR4, was significantly expressed (FPKM > 0.5) in all cell types, and had FPKM values of 7.91 (astrocytes), 1.10 (neurons), 7.92 (oligodendrocytes), and

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Table 2. Similarity score and docking score of Mavorixafor (reference) and five selected compounds.

Compound	Formula	Similarity	Docking score	Residues
Mavorixafor (reference)	$C_{21}H_{27}N_5$		8.3	TYR45, AL112, ILE185, ASP18, GLU288
ZINC49069258	$C_{22}H_{25}N_5$	0.77	8.5	TRP94, ASP97, TRP102, VAL112, TYR116, TYR25, GLU288
ZINC49067615	$C_{21}H_{25}N_5$	0.72	8.5	LEU41, TRP94, ASP97, VAL112, ARG183
ZINC49069264	$C_{21}H_{25}N_5O$	0.70	8.5	TRP94, ASP97, ALA98, TYR116, ASP187, HIS113
ZINC103242147	$C_{23}H_{32}N_4$	0.61	8.9	LEU41, TRP94, ASP97, TRP102, VAL112, TYR116, CYS18, HIS113
ZINC1353043237	$C_{23}H_{32}N_4$	0.56	8.6	LEU41, TYR45, TRP94, ASP97, ALA98, TRP102, VAL112, TRY11, GLU288

109.80 (microglia). It was also upregulated in both AD and PD. Thus, CXCR4 was selected for the next part of the study. Previous studies have reported the upregulation of CXCR4 in AD [59] and PD [60], which is in line with our results. For example, it was found out that CXCR4/CXCL12-based mechanism suggests that Aβ plaques attract microglia to activate the inflammatory cascade by which CXCL12 stimulates CXCR4-dependent signaling both in microglia and in astrocytes to release pro-inflammatory cytokines such as tumor necrosis factor α. The involvement of Ca^{2+} cascade is also suggested to be involved based on this signaling mechanism, which ends in activation of kinases, phosphorylation, and further excitotoxicity cascade triggered by excessive stimulation by glutamate [61]. Likewise, Tian *et al.* demonstrated that HMGB1 A Box protects TH+ neurons by binding CXCR4 to inhibit the migration/infiltration of T cells and macrophages to substantia nigra mediated by HMGB1/CXCL12 complex formed by neuron-derived HMGB1, and thus prevents neuronal damage in PD mice model [62].

Molecular docking

Virtual screening was performed via molecular docking to screen novel potential therapeutic agents, by computing their binding affinity score and interacting residues of our target protein, CXCR4. 3D structure of selected protein CXCR4 (PDB ID: 3ODU) was retrieved from Protein Data Bank. Mavorixafor was selected as a reference drug as it is a known inhibitor of CXCR4 and can pass through the BBB. Its 3D structure was retrieved from PubChem. The ZINC database was searched to find molecules showing at least 40% similarity with Mavorixafor to find molecules that could have similar biological activity. Fifty-two such molecules were selected as test molecules to perform molecular docking. Mavorixafor and these test molecules were docked with the CXCR4 (PDB ID: 3ODU) using AutoDock Vina. Mavorixafor, our reference drug, displayed docking score or binding affinity (ΔG) of −8.3 kcal/mol. Among our 52 test molecules, 5 molecules were selected that displayed higher docking score or binding affinity (ΔG) than that of Mavorixafor. These molecules were ZINC103242147 (−8.9 kcal/mol), ZINC1353043237 (−8.6 kcal/mol), ZINC49069258 (−8.5 kcal/mol), ZINC49067615 (−8.5 kcal/mol), and ZINC49069264 (−8.5 kcal/mol). This indicates that these molecules could show higher inhibitory activity toward CXCR4 as compared to Mavorixafor. The binding affinity scores and interacting residues for reference and all selected molecules have been listed in Table 2.

ADME analysis

Mavorixafor and all five test molecules were predicted to be orally bioavailable (Fig. 3). The radar plot of all six compounds falls in the pink zone and thus can be deemed to be drug-like, satisfying all six criteria: XLOGP3 between −0.7 and +5.0, molecular weight between 150 and 500 g/mol, TPSA between 20 and 130 Å, log S not higher than 6, fraction of carbons in the sp3 hybridization not less than 0.25, and no more than 9 rotatable bonds. The molecular and physicochemical properties of Mavorixafor and five test molecules were evaluated (Table 3). Mavorixafor and five test molecules were tested for the five predictors of lipophilicity (iLOGP, XLOGP3, WLOGP, MLOGP, Silicos-IT log P) and the consensus log $P_{o/w}$ (Table 3). The water solubility of Mavorixafor and five test molecules was analyzed by three methods (Ali, ESOL, and SILICOS-IT). The resulting log S values (where S is molar solubility in water) and respective qualitative solubility classes have been mentioned in Table 3. BBB permeability, gastrointestinal absorption, skin permeation, and effect on permeability glycoprotein (Pgp) and major isoforms of cytochromes P450 (CYP) were predicted for all molecules (Table 3). To evaluate the drug-likeness of all molecules were tested by Lipinski, Ghose, Veber, Egan, and Muegge methods (Table 3). No molecule exhibited any violation in any of the five methods. The bioavailability score of all molecules was calculated to be 0.55 for all molecules. No undesirable moieties were observed under the PAINS filter for any molecule. Besides ZINC49069258, no other molecule exhibited undesirable moieties under the Brenk filter. ZINC49067615, ZINC49069264, and ZINC103242147 showed violations related to lead-likeness (Table 3).

MD simulation

Five test molecule complexes with CXCR4 displaying the highest docking scores were chosen for MD simulations. The RMSD values offered an understanding of these complexes' stability, with the docked structure values lying between 0.15 nm and 0.45 nm (Fig. 4A). A low fluctuation in the RMSD values implies that inhibitors achieved stability in the docked structure. RMSF values per residue for all molecules are mostly less than 0.4 nm (Fig. 4B), indicating the stability of complexes. In addition, the elastic stability of the studied complexes was examined as radius of gyration (Rg) values varied between 2.1 nm and 2.4 nm, indicating that the system was able to achieve stability after 15 ns (Fig. 4C). ZINC103242147 showed a stronger Coulomb potential of interaction (Fig. 4D), while ZINC49067615 has a stronger Lennard–Jones potential of interaction energy (Fig. 4E). This suggests that these two molecules could show more significant inhibitory activity than Mavorixafor. The molecular dynamics trajectory was divided into 10-ns intervals to analyze the minimum distance and interactions between protein residues and ligand molecules. The first 10 ns was excluded as the duration needed for the system to reach stability. The minimum distance was calculated using the 'gmx mindist' command of gromacs, and residues with an average distance of 0.35 nm or less have been shown in Fig. 5. The heatmap

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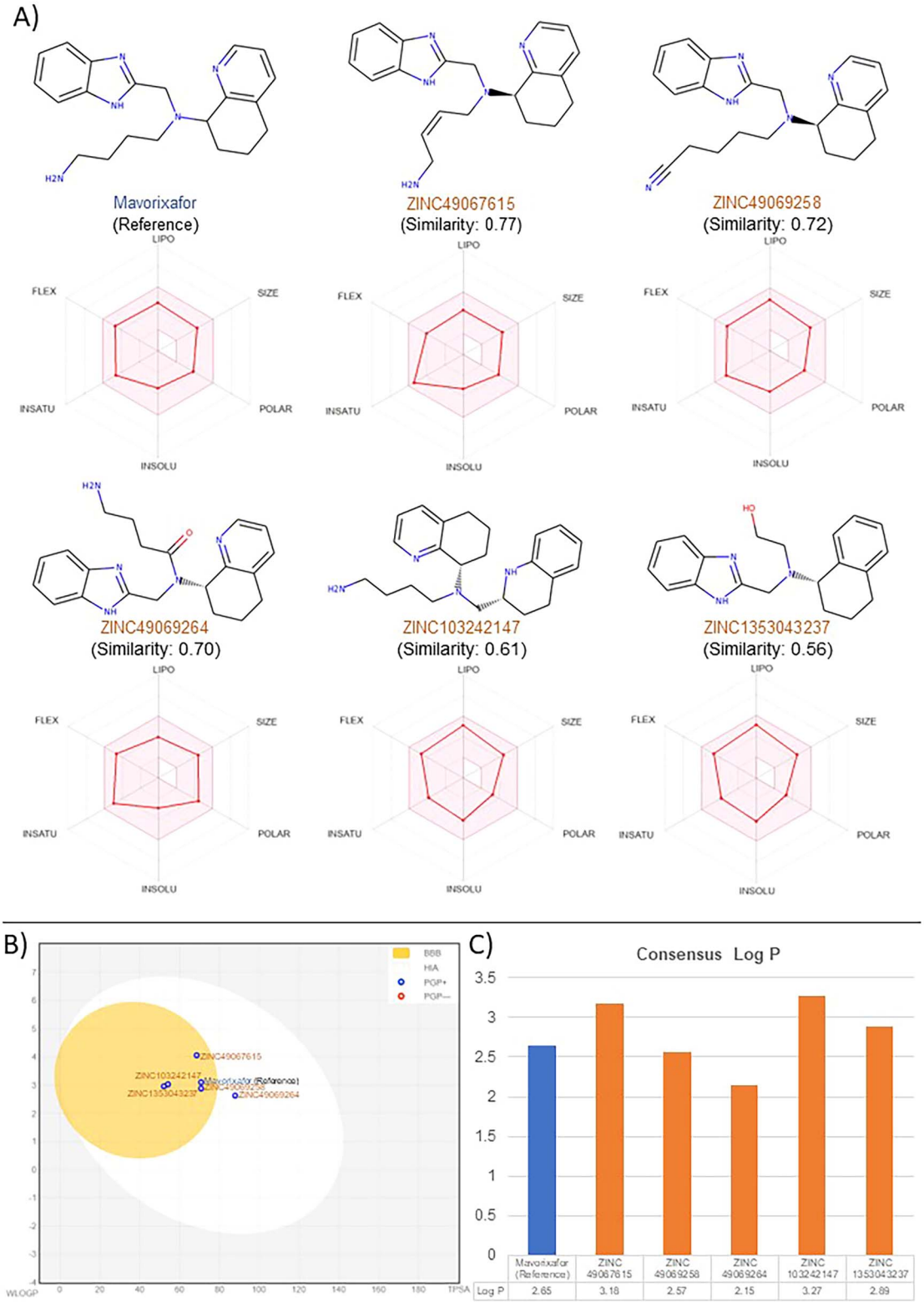


Figure 3. ADME analysis of Mavorixafor (reference drug) and five test compounds. (A) Molecular structure, bioavailability radar, and similarity score with the reference drug. The hexagonal zone denotes the optimal range for each property (lipophilicity: XLOGP3 b/w −0.7 and +5.0, size: molecular weight b/w 150 and 500 g/mol, polarity: TPSA b/w 20 and 130 Å, solubility: log S not higher than 6, saturation: fraction of carbons in the sp3 hybridization not less than 0.25, and flexibility: no more than 9 rotatable bonds. (B) BOILED-Egg prediction of blood–brain barrier penetration and gastrointestinal absorption. (C) SwissADME calculates five predictors (iLOGP, MLOGP, WLOGP, XLOGP3, and SILICOS-IT) to compute the consensus log $P_{o/w}$ (mean of all five predictors).

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Table 3. Different descriptors for ADME properties of Mavorixafor and 5 test compounds.

Descriptors		Mavorixafor (reference)	ZINC49067615	ZINC49069258	ZINC49069264	ZINC103242147	ZINC1353043237
Physicochemical properties	Formula	$C_{21}H_{27}N_5$	$C_{22}H_{25}N_5$	$C_{21}H_{25}N_5$	$C_{21}H_{25}N_5O$	$C_{23}H_{32}N_4$	$C_{20}H_{23}N_3O$
	MW	349.47	359.47	347.46	363.46	364.53	321.42
	#Heavy atoms	26	27	26	27	27	24
	#Aromatic heavy atoms	15	15	15	15	12	15
	Fraction Csp3	0.43	0.41	0.33	0.38	0.52	0.35
	#Rotatable bonds	7	7	6	7	7	5
	#H-bond acceptors	4	4	4	4	3	3
	#H-bond donors	2	1	2	2	2	2
	MR	105.45	107.29	104.97	105.65	115.54	96.49
	TPSA	70.83	68.6	70.83	87.9	54.18	52.15
Lipophilicity	iLOGP	2.2	2.33	2.67	2.04	3.54	2.3
	XLOGP3	2.38	3.01	1.97	1.41	3.42	2.95
	WLOGP	3.1	4.06	2.88	2.63	3.03	2.96
	MLOGP	1.91	2.06	1.84	1.43	2.66	2.52
	Silicos-IT log P	3.68	4.44	3.5	3.22	3.71	3.72
	Consensus log P	2.65	3.18	2.57	2.15	3.27	2.89
Water solubility	ESOL log S	−3.47	−3.91	−3.27	−2.93	−4.12	−3.82
	ESOL solubility (mg/ml)	1.18E−01	4.38E−02	1.88E−01	4.26E−01	2.76E−02	4.82E−02
	ESOL solubility (mol/l)	3.38E−04	1.22E−04	5.42E−04	1.17E−03	7.56E−05	1.50E−04
	ESOL class	Soluble	Soluble	Soluble	Soluble	Moderately soluble	Soluble
	Ali log S	−3.51	−4.12	−3.08	−2.86	−4.24	−3.71
	Ali solubility (mg/ml)	1.08E−01	2.76E−02	2.87E−01	5.01E−01	2.11E−02	6.30E−02
	Ali solubility (mol/l)	3.10E−04	7.67E−05	8.26E−04	1.38E−03	5.78E−05	1.96E−04
	Ali class	Soluble	Moderately soluble	Soluble	Soluble	Moderately soluble	Soluble
	Silicos-IT log Sw	−6.9	−7.33	−6.18	−6.43	−7.02	−6.26
	Silicos-IT solubility (mg/ml)	4.44E−05	1.67E−05	2.31E−04	1.35E−04	3.46E−05	1.77E−04
	Silicos-IT solubility (mol/l)	1.27E−07	4.63E−08	6.64E−07	3.72E−07	9.48E−08	5.50E−07
	Silicos-IT class	Poorly soluble	Poorly soluble	Poorly soluble	Poorly soluble	Poorly soluble	Poorly soluble
Pharmacokinetics	GI absorption	High	High	High	High	High	High
	BBB permeant	Yes	Yes	Yes	No	Yes	Yes
	Pgp substrate	Yes	Yes	Yes	Yes	Yes	Yes
	CYP1A2 inhibitor	Yes	Yes	No	No	No	Yes
	CYP2C19 inhibitor	Yes	Yes	Yes	Yes	No	No
	CYP2C9 inhibitor	No	Yes	No	No	No	No
	CYP2D6 inhibitor	Yes	Yes	Yes	Yes	Yes	Yes
	CYP3A4 inhibitor	Yes	Yes	Yes	Yes	Yes	Yes
	Log K_p (cm/s)	−6.74	−6.36	−7.02	−7.52	−6.1	−6.17
Drug-likeness	Lipinski #violations	0	0	0	0	0	0
	Ghose #violations	0	0	0	0	0	0
	Veber #violations	0	0	0	0	0	0
	Egan #violations	0	0	0	0	0	0
	Muegge #violations	0	0	0	0	0	0
	Bioavailability score	0.55	0.55	0.55	0.55	0.55	0.55
Medicinal chemistry friendliness	PAINS #alerts	0	0	0	0	0	0
	Brenk #alerts	0	0	1	0	0	0
	Lead-likeness #violations	0	1	0	1	1	0
	Synthetic accessibility	3.38	3.36	3.45	3.31	3.99	2.91

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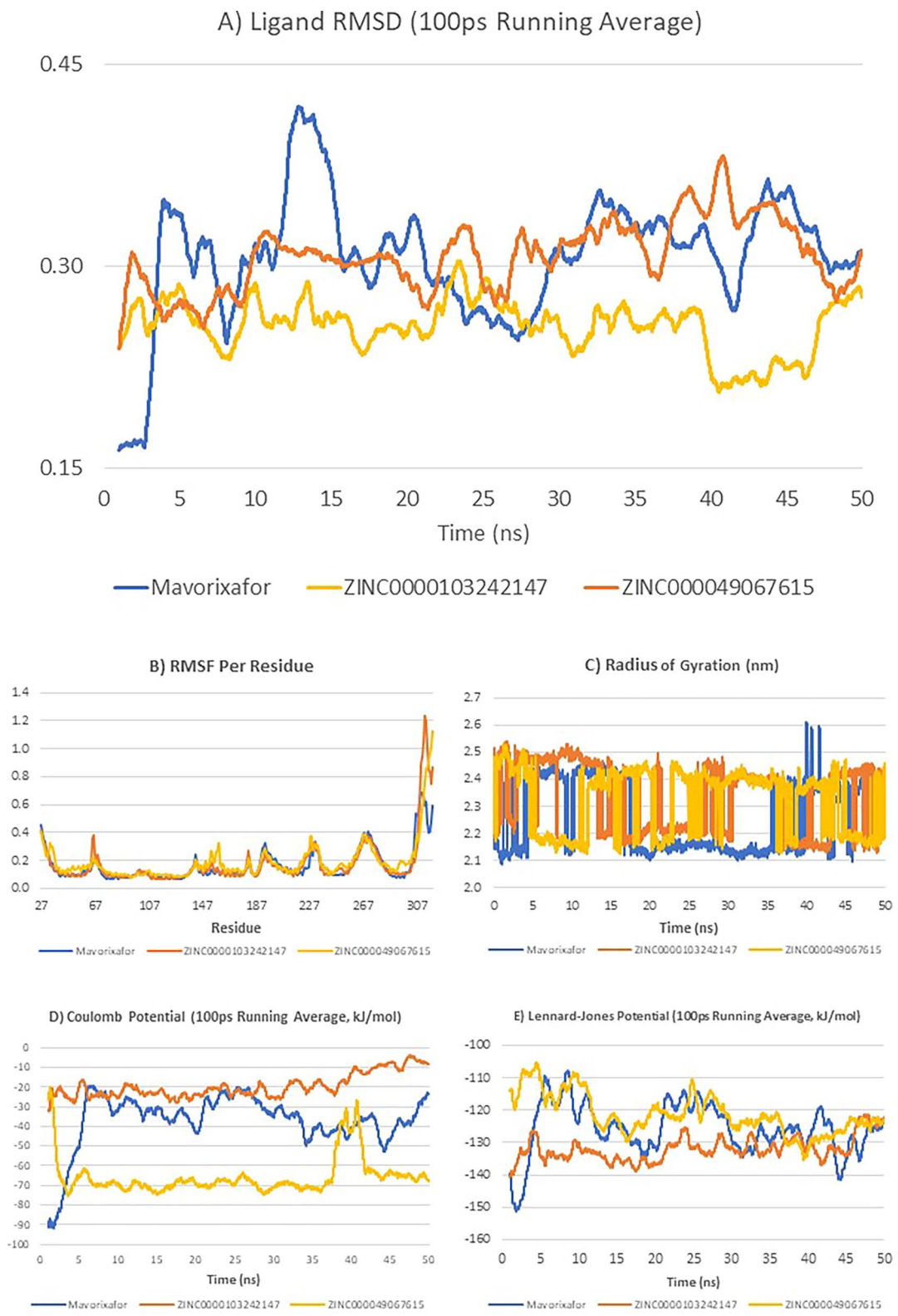


Figure 4. Results of molecular dynamics simulation. (A) RMSD of ligand in terms of 100 ps running averages. The RMSD values of these complexes lied between 0.15 nm and 0.45 nm. A low fluctuation in the RMSD values implies that inhibitors achieved stability in the docked structure. (B) Root-mean-square fluctuation (RMSF) per residue. RMSF values per residue for all molecules are mostly less than 0.4 nm, indicating the stability of complexes. (C) Radius of gyration of the protein–ligand complexes. The elastic stability of the studied complexes was examined as radius of gyration (Rg) values varied between 2.1 nm and 2.4 nm, indicating that the system was able to achieve stability after 15 ns. (D) Interaction energy between protein and ligand—Coulomb potential. ZINC103242147 showed a stronger Coulomb potential of interaction. (E) Interaction energy between protein and ligand—Lennard–Jones potential. ZINC49067615 has a stronger Lennard–Jones potential of interaction energy.

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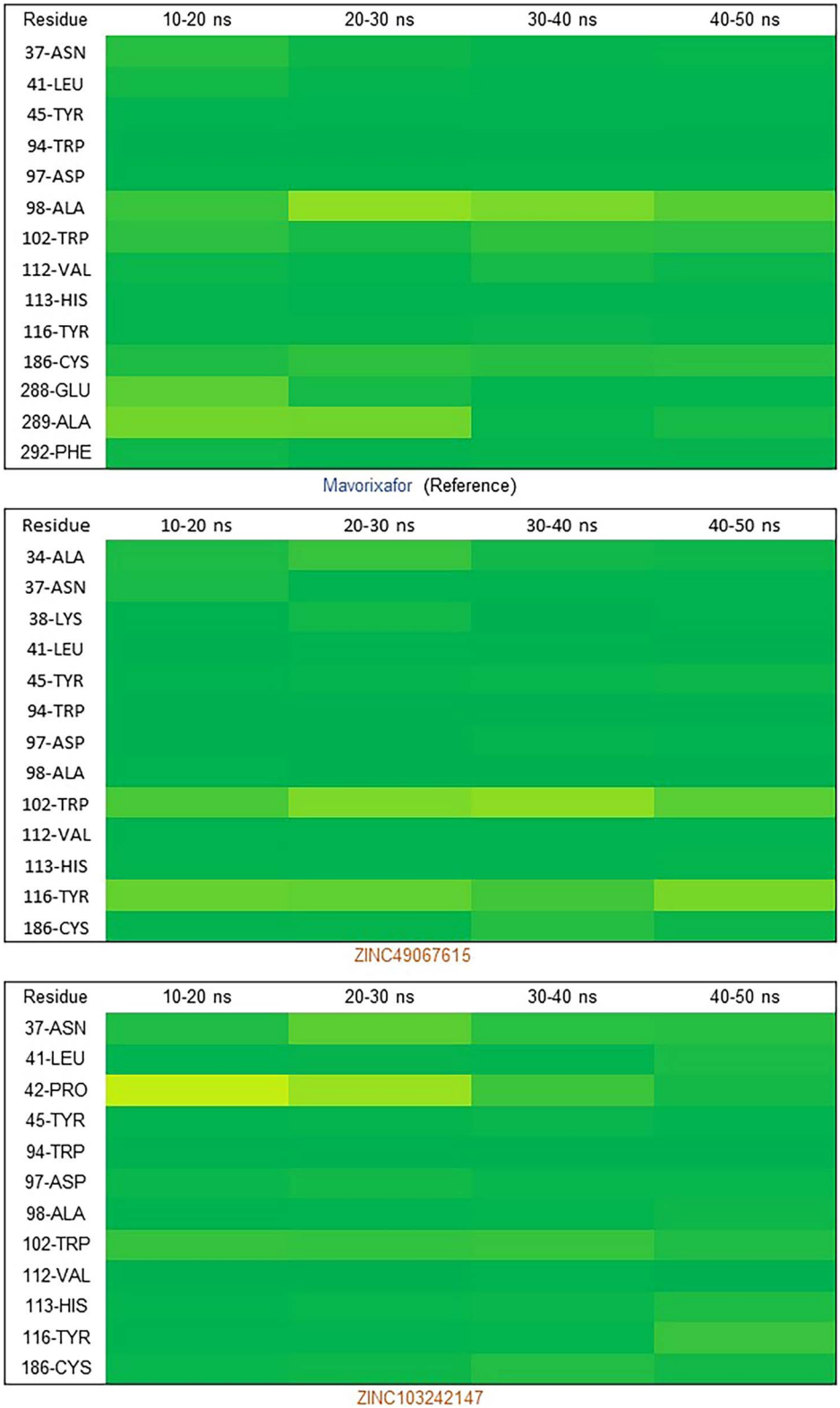


Figure 5. Minimum distance between ligand and protein residues. The minimum distance was calculated using the 'gmx mindist' command of gromacs, and residues with an average distance of 0.35 nm or less have been shown. The first 10 ns of the simulation were excluded as the duration needed for the system to reach stability. The simulation was divided into 10-ns intervals, and a heatmap was generated by showing the amount of time distance was less than 0.35 nm. Green signifies that the distance was less than 0.35 nm for 100% of the duration, while yellow indicates that the distance was less than 0.35 nm for 0% of the duration. Specific residues like TYR45, TRP94, and ASP97 can be seen to be present for most of the duration of the simulation.

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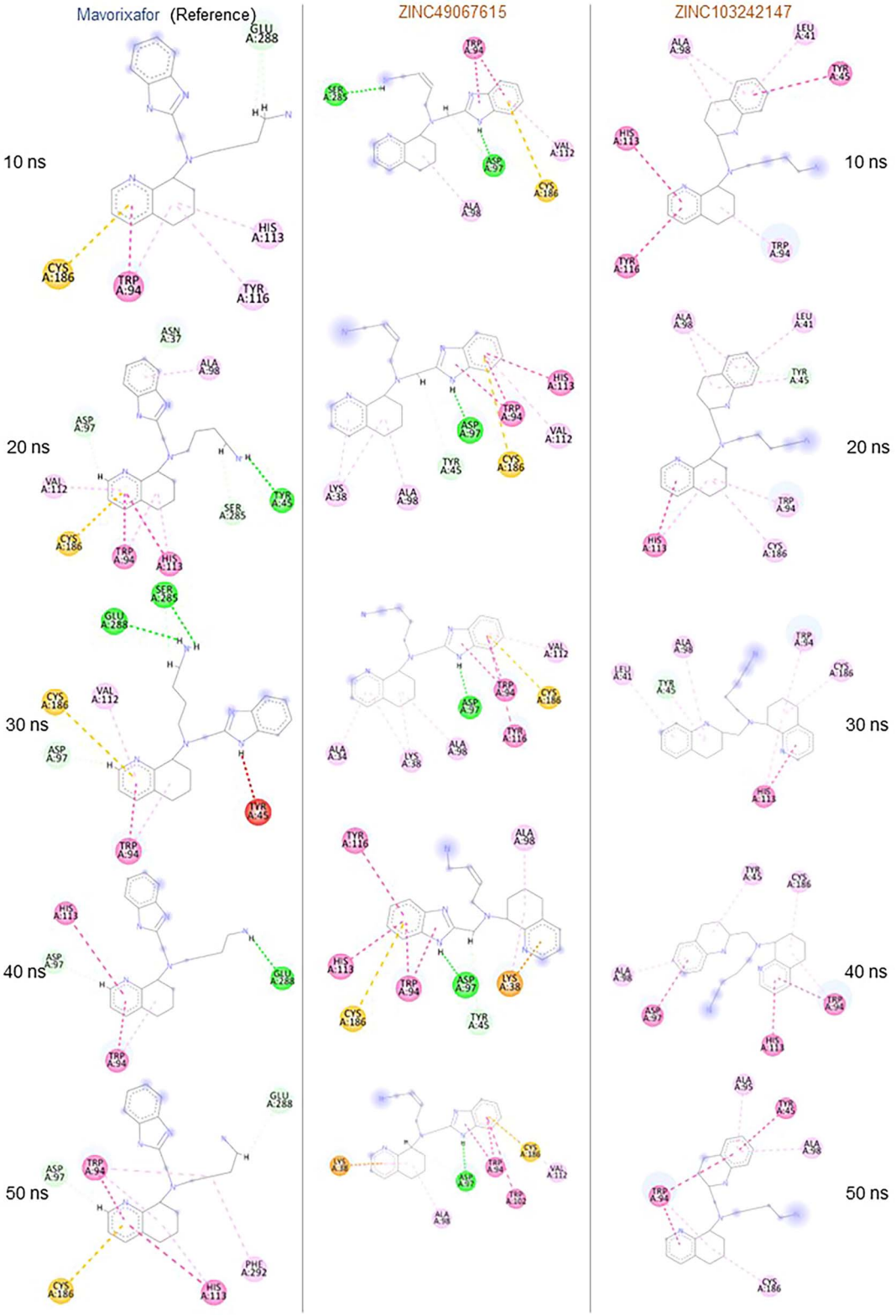


Figure 6. Interaction between ligand and protein residues at 10 ns, 20 ns, 30 ns, 40 ns, and 50 ns. The first 10 ns of the simulation were excluded as the duration needed for the system to reach stability. Certain residues like TYR45, TRP94, and ASP97 can be seen to be present throughout the different time points (10 ns, 20 ns, 30 ns, 40 ns, and 50 ns) of the simulation.

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Table 4. Neurodegenerative diseases with their diagnostic targets and respective drugs that have been identified using Computer-aided drug discovery (the experimental validation, either *in vitro* or *in vivo*, performed for each drug is also indicated).

Disease	Targets	Molecules	Software	*In vitro* or *in vivo* study
Alzheimer's disease	N-Methyl-D-aspartate (NMDA) receptor	1-Benzyl-1,2,3,4-tetrahydro-β-carboline [80]	ICM	Cell-based assay [80]
	Acetylcholinesterase (AChE) and butyrylcholinesterase (BuChE)	6-Chloro-pyridonepezils [81]	Autodock Vina, QikProp	*In vitro* blood–brain barrier (BBB) model [92]
	Nucleoside hydrolase	Flavonoids [82]	AutoDock	Mice and rat models [93, 94]
	N-Methyl-D-aspartate (NMDA) receptor	Ifenprodil [83]	Schrödinger Suite	Chicken embryo forebrain cultures (E10) [83]
	AChE, BuChE, BACE 1, MAO, and NMDA	Memantine [84, 85]	Glide	Clinical trial [95]
	AChE	Morin [86]	Glide	APPswe/PS1dE9 mice [96]
	ABCG2 enzyme	2,4-Disubstituted pyridopyrimidine derivatives [87]	Autogrid, Autodock, and GROMACS	*In vitro* enzyme inhibitory model [97]
	AChE	Pyridonepezil [88]	Autodock Vina	*In vitro* blood–brain barrier model [92]
	Transient receptor potential canonical 6 (TRPC6)	Piperazine derivatives [89]	PASS software	Ellman's method [98]
	Human islet amyloid polypeptide (hIAPP)	Rutin [90]	AutoDock and AutoDock Vina	Neuroblastoma cells (IMR32) and Wistar rats [99]
Parkinson's disease	Leucine-rich repeat kinase 2 (LRRK2)	9-Methyl-*N*-phenylpurine-2,8-diamine, *N*-phenylquinazolin-4-amine, and 1,3-dihydroindol-2-one [91]	MOE	*In vitro* and *in vivo* studies [100]

was generated for residues showing the amount of time distance was less than 0.35 nm. The green color signifies that the distance was less than 0.35 nm for 100% of the duration, while yellow indicates that the distance was less than 0.35 nm for 0% of the duration. Specific residues like TYR45, TRP94, and ASP97 can be seen to be present for most of the duration of the simulation. Interaction between ligand and protein residues at 10 ns, 20 ns, 30 ns, 40 ns, and 50 ns has been shown in Fig. 6. Here again, specific residues like TYR45, TRP94, and ASP97 can be seen to be interacting with ligands at different timeframes of the simulation, signifying the importance of these residues for receptor–ligand interaction.

DISCUSSION

AD pathophysiology is characterized by abnormal Aβ aggregation and tau protein hyperphosphorylation [63], while PD pathophysiology is characterized by abnormal α-Syn [64]. Inhibition of accumulation or promoting clearance Aβ and α-Syn are potential therapeutic targets for treatment of AD and PD, respectively [65, 66]. CXCR4 is a receptor present on the surface of B lymphocytes, which secretes antibodies that target toxic proteins in AD and PD [67]. Downregulation of CXCR4 is essential for the migration of immature B cells into the peripheral blood from the bone marrow [68]. CXCR4 upregulation in AD patients results in decreased migration of B cells and decline in anti-Aβ antibody secretion, leading to Aβ aggregation [69]. Microarray analysis has revealed upregulation of CXCR4 in the inferior parietal lobe brain sections of AD patients, which was also confirmed by real-time PCR and immunohistochemistry [70]. AMD3100, a CXCR4 inhibitor, has been shown to improve cognitive performance, reduce neuroinflammation, and alleviate AD pathophysiology [71]. However, normal levels of CXCR4 expression is required for cognitive performance and normal development of hippocampal dentate gyrus [72]. Chronic administration of AMD3100 has been shown to result in learning and memory dysfunction in young mice [73]. Thus, the crux of the treatment lies in maintaining CXCR4 levels. In another study, MPTP-mediated CXCR4 upregulation preceded the loss of DA neurons, suggesting role of CXCR4 in the PD etiology and its potential as a new target molecule for PD treatment [74]. AMD3100 also increased life span and improved motor function in the $SOD1^{G93A}$ mice model of ALS [75]. Likewise, Mason *et al.* demonstrated that AMD3100 pre-treated animals exhibited diminutive AMPH-induced locomotor activity compared to SAL pre-treated animals. Postmortem analyses of brain tissue revealed elevated CXCR4 protein levels in the striatum of all experimental groups. The results implicated that CXCR4 inhibition with AMD3100 attenuates amphetamine-induced locomotor activity [76]. Furthermore, CXCR4 knockout prevents degeneration of dopamine neurons of MPTP-lesioned mice through microglial and astroglial activation. CXCR4 knockout also protected the BBB from MPTP-induced damage [77]. Therefore, downregulation of CXCR4 can be a promising approach for the treatment for AD and PD.

However, drug discovery is an expensive and time-consuming process with the average approved drug requiring 10 to 15 years to develop with an estimated cost of 0.8–2 billion USD. Many licensed drugs, such as captopril, dorzolamide, oseltamivir, aliskiren, and nolatrexed, were all optimized using CADD [78], and a large number of publications describe the successful design and discovery of leads/drugs using CADD [79]. The drugs that were optimized

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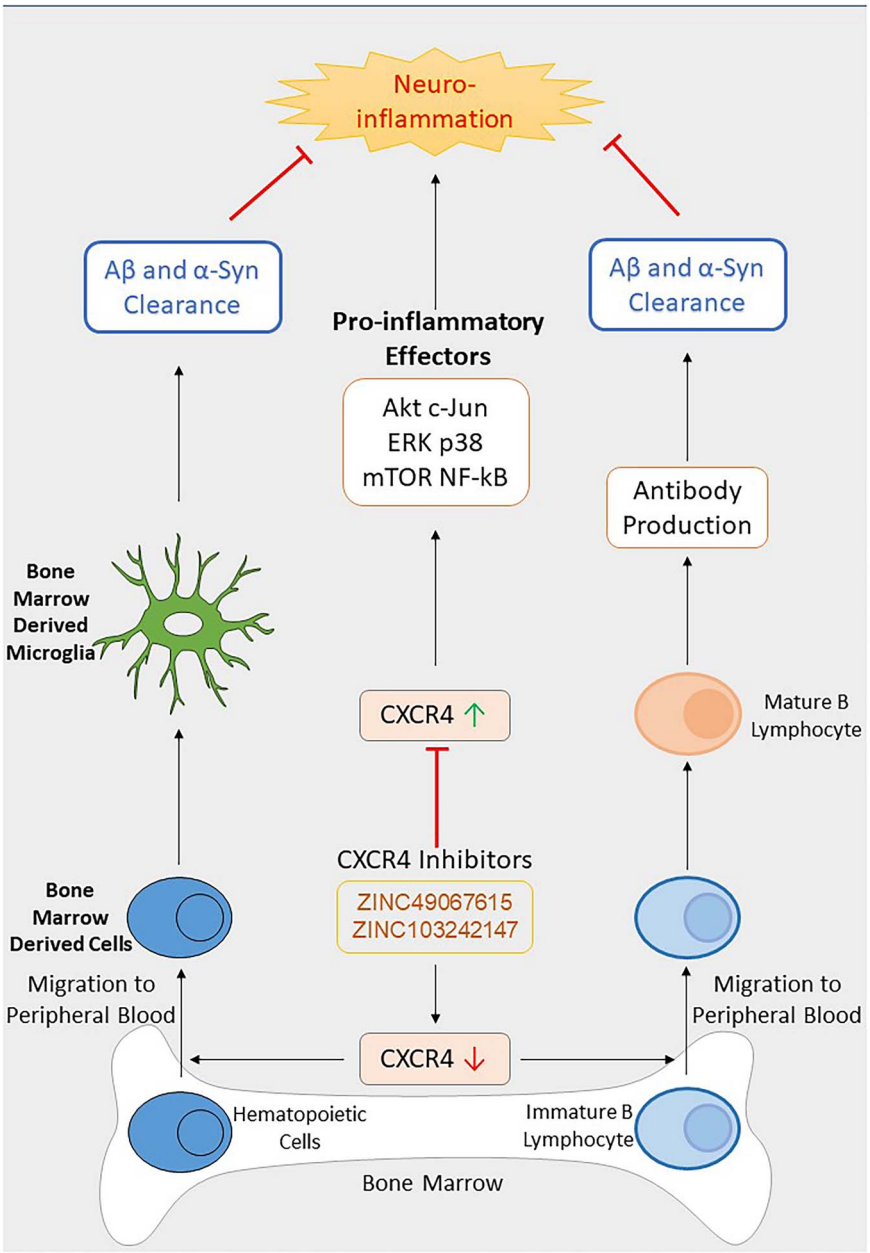


Figure 7. Downregulation of CXCR4 is essential for the migration of immature B lymphocytes into the peripheral blood from the bone marrow. After maturation, these B lymphocytes secrete antibodies that target toxic proteins such as Aβ and α-Syn. Thus, downregulation of CXCR4 can help in Aβ and α-Syn clearance. Conversely, upregulation of CXCR4, being a pro-inflammatory chemokine, activates effectors (such as Akt, c-Jun, ERK, p38, mTOR, and NF-kB) that leads to neuroinflammation, a hallmark of neurodegenerative disease, such as AD and PD. CXCR4 inhibitors can reduce its expression level to normal range, thereby reducing neuroinflammation and providing neuroprotection.

or designed [80–91] for AD or PD and later confirmed through *in vitro* or *in vivo* studies [92–100] have been listed in Table 4. The main goal of CADD is to reduce these timescales and costs without affecting quality [101]. Importantly, CADD can be used in most stages of drug development: from target identification to target validation, from lead discovery to optimization, and in preclinical studies. It is therefore estimated that CADD could reduce the cost of drug development by up to 50% [102, 103]. Therefore, we our study aims to find potential leads for the treatment of AD and PD among the large number of small molecule compounds available in online databases.

We searched for molecules similar to Mavorixafor, a known CXCR4 inhibitor, in ZINC Database and found 52 such compounds. Due to similarity in structure, these compounds may have similar bioactivity, which we later validated using molecular docking and molecular dynamics simulation. Through molecular docking, we found five molecules that had better binding affinity than Mavorixafor. The prediction of the fate of a drug and the effects caused by a drug inside the body, such as how much drug is absorbed if administered orally and how much is absorbed in the gastrointestinal tract, are an indispensable part of drug discovery. In a similar way, if the absorption is poor, its distribution and metabolism would be affected, which can lead to neurotoxicity and nephrotoxicity. Ultimately, the study aims to understand the disposition of a drug molecule within an organism. Thus, ADME study is one of the most essential

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parts of computational drug design. Therefore, we validated the molecules through their ADME profile and found that four out of five of these molecules were suitable as lead compounds, the only exception being ZINC49069264, which was not BBB permeable. Then, we employed MD simulation to validate stability of protein–ligand complexes. We found two molecules, ZINC49067615 and ZINC103242147, that were stable in their interaction with CXCR4 and had stronger binding energies than Mavorixafor. Certain residues such as TYR45, TRP94, and ASP97 were found to be involved in binding between these molecules and CXCR4 through molecular docking as well as MD simulation. This is in line with the residues that have been reported to be important for stability of interactions with known CXCR4 inhibitors [104, 105], further confirming our results.

CONCLUSION

The molecular docking and simulation were used to screen the molecules showing structural similarity with Mavorixafor (CXCR4 inhibitor) to find novel molecules potentially regulating the CXCL12/CXCR4 pathway. After the virtual screening of the molecules using molecular docking, five test molecules exhibiting high docking scores were selected for further analysis. After ADME analysis, ZINC49069264 was ruled out as it was observed to be unable to pass through the BBB. All other molecules were selected for further research. After MD simulation, ZINC49067615 and ZINC103242147 were stable in docked configuration with CXCR4 and exhibited high interaction energy. The ADME profiles of these compounds were also suitable for being used as lead compounds, passing all five, Lipinski, Ghose, Veber, Egan, and Muegge, descriptors of drug-likeness with no violations. Thus, this preliminary study revealed that these molecules can be potential therapeutic agents and be further tested in biological studies, either *in vitro* or *in vivo*, to analyze their potential in the modulation of the CXCL12/CXCR4 pathway to treat AD and PD (Fig. 7). However, the study accompanies the limitations, such as *in vitro* and *in vivo* validation of the identified compounds in respect to AD and PD pathology.

ACKNOWLEDGEMENTS

We would like to thank the senior management of Delhi Technological University (DTU) and the Department of Biotechnology (DBT), India's Government, for their constant support and financial assistance.

CONFLICT OF INTEREST STATEMENT

None declared.

DATA AVAILABILITY

RNA-seq data used for this study is present in the Gene Expression Omnibus database under accession numbers GSE67333 and GSE114517, available at the following URLs: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE67333 and https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE114517.

AUTHORS' CONTRIBUTIONS

PK and RT conceived and designed the manuscript. RT has collected, analyzed, and critically evaluated these data. PK and RT have prepared figures and tables. PK and RT analyzed the entire data and wrote the manuscript.

ABBREVIATIONS

CXCR4: C-X-C chemokine receptor type 4; CXCL12: C-X-C motif chemokine 12; NDD: neurodegenerative disorders; AD: Alzheimer's disease; LOAD: late-onset Alzheimer's disease; NFT: neurofibrillary tangles; PD: Parkinson's disease; ALS: amyotrophic lateral sclerosis; Aβ: amyloid-β; α-Syn: α-synuclein; CNS: central nervous system; GEO: Gene Expression Omnibus; GREIN: GEO RNA-seq Experiments Interactive Navigator; DEGs: differentially expressed genes; PPI: protein–protein interaction; BMP4: bone morphogenetic protein 4; CDKN2B: cyclin dependent kinase inhibitor 2B; CXCR4: C-X-C motif chemokine receptor 4; GNRH1: gonadotropin-releasing hormone 1; GRM2: glutamate metabotropic receptor 2; KCNMB3: potassium calcium-activated channel subfamily M regulatory beta subunit 3; SLC13A4: solute carrier family 13 member 4; SLC26A4: solute carrier family 26 member 4; SLC5A5: solute carrier family 5 member 5; C1QL3: complement C1q like 3; DCC: DCC netrin 1 receptor; FST: follistatin; GNLY: granulysin; GPR26: G protein-coupled receptor 26; NLRP2: NLR family pyrin domain containing 2; NR4A2: nuclear receptor subfamily 4 group A member; NRIP3: nuclear receptor interacting protein 3; OXTR: oxytocin receptor; RAB3C: member RAS oncogene family; SLC18A2: solute carrier family 18 member A2; TRPC4: transient receptor potential cation channel subfamily C member 4; UNC45B: Unc-45 myosin chaperone B; WDR63: WD repeat domain 63; MD: molecular dynamics; Rg: radius of gyration; TYR: tyrosine; TRP: tryptophan; ASP: aspartate; GPCR: G protein-coupled receptor; ADME: absorption, distribution, metabolism and excretion; PDB: Protein Data Bank; CYP: cytochrome P450; PAINS: pan assay interference compounds; RMSD: root-mean-square deviations; RMSF: root-mean-square fluctuation; p53 protein: cellular tumor antigen p53; Rb protein: retinoblastoma protein

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